Personalized Travel Itinerary Planner: AI-Powered Itinerary Generator

Date	20 October 2026
Team ID	Individual project (no team)
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	-

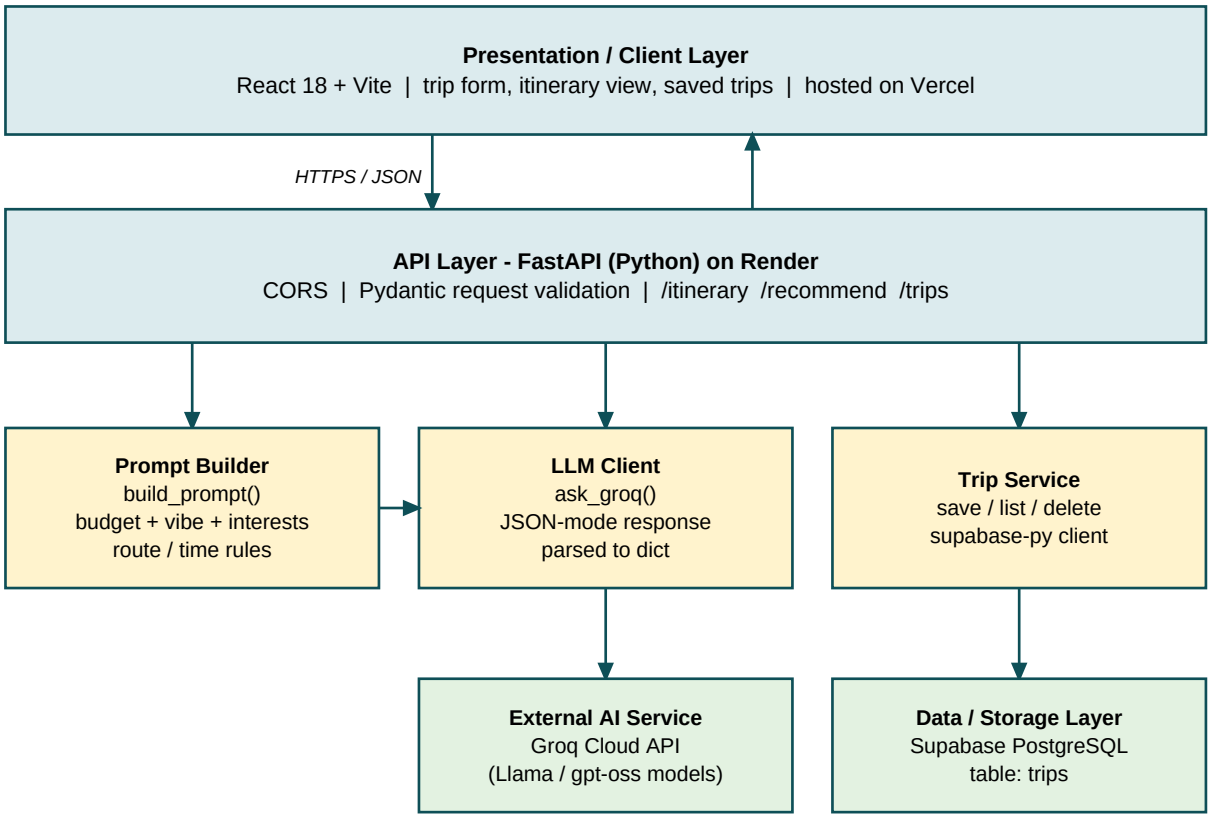
Project Description:
The Personalized Travel Itinerary Planner is a full-stack, AI-powered web application that turns a traveller's destination, number of days, budget, vibe and interests into a clear day-by-day itinerary. A React frontend collects the preferences, a FastAPI backend builds a structured prompt and calls the Groq LLM in JSON mode, and Supabase stores saved trips. The planner includes timings, travel times, cost estimates, dining and stay tips, destination suggestions, save / view / delete, and copy-as-text. The frontend is deployed on Vercel and the backend on Render.

Scenarios

Scenario 1 - Solo traveller: enters Goa, 3 days, medium budget, interests food and beaches, and gets a balanced plan with nearby places grouped and affordable dining.
Scenario 2 - Couple: picks the romantic vibe and a premium budget, and gets a relaxed, scenic plan with estimated costs.
Scenario 3 - Group of friends: combines adventure and culture interests; group planning is part of the future roadmap.

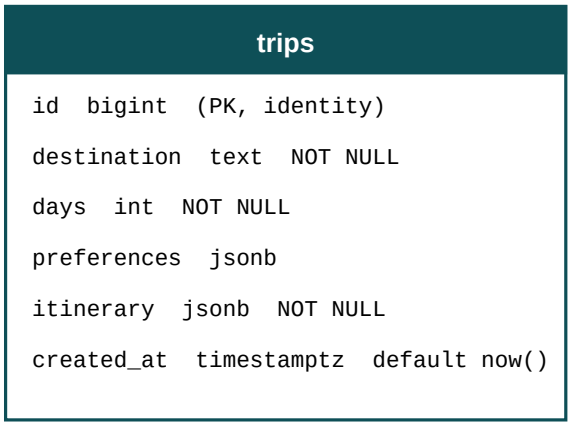
Technical Architecture

A three-tier architecture: React + Vite presentation layer (Vercel), FastAPI application layer with prompt builder, LLM client and trip service (Render), and Supabase PostgreSQL data layer. The Groq API is the external AI service.



ER Diagram (database)

The application uses one table, trips. Preferences and the generated itinerary are stored as JSONB so any itinerary shape can be saved without schema changes.



Pre-requisites

- Python 3 and FastAPI basics (https://fastapi.tiangolo.com)
- React and Vite basics (https://react.dev, https://vitejs.dev)
- Groq API key (https://console.groq.com)
- Supabase account (https://supabase.com)
- Git and GitHub
- Render and Vercel accounts for deployment

Project Workflow

- Milestone 1: Environment setup (Epic 1) - tools, GitHub repository, folder structure
- Milestone 2: Supabase database setup (Epic 2) - project, trips table, environment variables
- Milestone 3: Preference collection and recommendation (Epic 3) - form, prompt structuring, destination suggestions
- Milestone 4: Itinerary generation and personalisation (Epic 4) - daily builder, route and time ordering, vibe
- Milestone 5: Productisation (Epic 5) - save and manage trips, export, configuration
- Milestone 6: Deployment (Epic 6) - Render backend, Vercel frontend
- Milestone 7: Final testing and documentation (Epic 7) - functional tests, database validation, outputs

Milestone 1: Environment Setup

Activity 1.1-1.3: Created a GitHub repository, set up VS Code with Python and Node.js, and used AI coding tools to generate the initial folder structure: backend/ (FastAPI) and frontend/ (React + Vite).

Milestone 2: Supabase Database Setup

Activity 2.1-2.3: Created a Supabase project, ran the SQL below to create the trips table, and stored the keys in a local .env file (excluded from Git). The same four variables were added in Render.

```
create table if not exists trips (
  id bigint generated always as identity primary key,
  destination text not null,
  days int not null,
  preferences jsonb,
  itinerary jsonb not null,
  created_at timestamptz default now()
);
```

```
GROQ_API_KEY=...   GROQ_MODEL=...   SUPABASE_URL=...   SUPABASE_KEY=...
```

Milestone 3: Preference Collection & Recommendation Engine

Activity 3.1: The React form captures destination, days, budget, vibe and interests. Activity 3.2: build_prompt() converts them into a structured prompt. Activity 3.3: POST /recommend returns five destination ideas for the chosen budget, interests and vibe.

Milestone 4: Itinerary Generation & Personalization

Activity 4.1-4.3: The prompt tells the model to match pace and activity style to the vibe, group nearby places on the same day, order activities by time, include travel time between stops, and return JSON in a fixed shape. ask_groq() calls the model in JSON mode and parses the result.

```
def ask_groq(prompt: str) -> dict:
    res = groq_client.chat.completions.create(
        model=MODEL,
        messages=[{"role": "system", "content": "You are an expert travel planner. Reply with valid JSON only."},
                  {"role": "user", "content": prompt}],
        response_format={"type": "json_object"},
        temperature=0.7)
    return json.loads(res.choices[0].message.content)
```

Milestone 5: Productization

Activity 5.1-5.4: POST /trips saves the preferences and itinerary, GET /trips lists them newest first, and DELETE /trips/{id} removes one. The frontend adds Save trip, View, Delete and Copy as text. Configuration values come from environment variables.

```
@app.post("/itinerary")
def itinerary(r: TripRequest):
    if not r.destination.strip():
        raise HTTPException(400, "Destination is required")
    return ask_groq(build_prompt(r))

@app.post("/trips")
def save_trip(t: SaveTrip):
    return supabase.table("trips").insert(t.model_dump()).execute().data[0]
```

Milestone 6: Deployment

Activity 6.1: Backend on Render - root directory backend, build command pip install -r requirements.txt, start command uvicorn main:app --host 0.0.0.0 --port $PORT, with the four environment variables.
Activity 6.2: Frontend on Vercel - root directory frontend, framework Vite, environment variable VITE_API_URL set to the Render URL.
Live application: https://travel-itinerary-planner-beryl.vercel.app
Backend docs: https://travel-itinerary-planner-ce17.onrender.com/docs

Milestone 7: Final Testing & Documentation

Activity 7.1: Functional testing of the full flow on the deployed site: build an itinerary, save it, check the Saved trips list. Activity 7.2: Database validation - the saved row appeared in the Supabase trips table with the preferences and itinerary JSON. A light performance test and the project documents are included in this repository.

Conclusion

The project delivers a working, deployed AI travel planner. The next phase adds user login, regenerating a single day, group trips and maps. Repository: https://github.com/niniwritescode/travel-itinerary-planner